

COURIER MANAGEMENT SYSTEM

PROJECT DOCUMENTATION

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A Web-Based Courier Management System

Academic Project Report

1. ABSTRACT

The Courier Management System is a web-based application developed to manage courier booking, shipment details, delivery status, customers, and delivery agents efficiently. The system allows customers to register, book courier services, enter sender and receiver details, and track their shipments.

The administrator can manage customers, courier bookings, delivery agents, shipments, and delivery status. The system reduces manual work, improves accuracy, saves time, and provides an easy way to track courier deliveries.

2. INTRODUCTION

Courier services play an important role in delivering documents, parcels, and other goods from one location to another. Managing courier services manually can be time-consuming and may lead to errors in maintaining customer and shipment information.

The Courier Management System provides a computerized solution for managing the complete courier process. It maintains customer information, courier booking details, shipment information, delivery-agent details, and tracking status in a centralized database.

3. PROBLEM STATEMENT

- Customer details are maintained manually.
- Courier booking takes more time.
- Tracking shipment status is difficult.
- Maintaining records is complicated.
- Manual data entry can cause errors.
- Generating reports is difficult.
- Customers may not receive timely status information.

4. OBJECTIVES

- To provide an easy courier booking system.
- To maintain customer details securely.
- To manage courier and shipment information.
- To provide shipment tracking.
- To manage delivery-agent information.
- To reduce manual work and paperwork.
- To improve accuracy and efficiency.
- To provide faster access to courier information.
- To maintain delivery status systematically.

- To generate useful reports for administrators.

5. EXISTING SYSTEM

In the existing system, courier activities may be handled manually or using separate records. Customer details, booking information, and delivery status are maintained using paper records or basic systems.

Disadvantages:

- More paperwork
- Time-consuming
- Difficult to search records
- Higher possibility of data-entry errors
- Difficult shipment tracking
- Difficult report generation
- Less efficient record management

6. PROPOSED SYSTEM

The proposed Courier Management System is a computerized system that manages courier operations through a single application. Customers can register and book courier services. The administrator can manage bookings, customers, delivery agents, and shipments. Shipment status can be updated throughout the delivery process.

Advantages:

- Easy courier booking
- Centralized database
- Faster record management
- Easy shipment tracking
- Reduced paperwork
- Improved accuracy
- Easy customer management
- Easy delivery-agent management
- Faster report generation

7. SCOPE OF THE PROJECT

- Customer registration and login
- Courier booking
- Sender and receiver management

- Shipment management
- Delivery-agent management
- Courier tracking
- Status updates
- Admin management
- Report generation

8. SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Hard Disk: 500 GB or above
- Keyboard and Mouse
- Internet connection

Software Requirements

- Operating System: Windows
- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Server: XAMPP
- Browser: Google Chrome / Microsoft Edge

9. MODULES

9.1 Admin Module

- Admin Login
- Manage Customers
- Manage Couriers
- Manage Delivery Agents
- Manage Shipments
- Update Delivery Status
- View Reports

9.2 Customer Module

- Registration
- Login
- Book Courier
- Enter Sender Details
- Enter Receiver Details
- View Booking Details
- Track Shipment

9.3 Delivery Agent Module

- Agent Login
- View Assigned Shipments
- View Customer Details
- Update Shipment Status
- Confirm Delivery

9.4 Shipment Module

This module manages Tracking ID, sender details, receiver details, parcel information, booking date, delivery agent, and shipment status.

9.5 Tracking Module

The tracking module allows customers and administrators to check the current shipment status.

Example: Booked → Picked Up → In Transit → Out for Delivery → Delivered

10. SYSTEM ARCHITECTURE

Customer / Admin / Delivery Agent → Web Application → PHP Backend → MySQL Database

The frontend provides the user interface, PHP handles application logic, and MySQL stores system data.

11. DATABASE DESIGN

Customer Table	Description
customer_id	Customer ID
name	Customer Name
email	Email
phone	Phone Number
address	Address
password	Login Password

Shipment Table	Description
shipment_id	Shipment ID
tracking_id	Tracking Number
customer_id	Customer ID
sender_name	Sender Name
receiver_name	Receiver Name
receiver_address	Receiver Address
parcel_type	Parcel Type
booking_date	Booking Date
status	Shipment Status

Delivery Agent Table	Description
agent_id	Agent ID
name	Agent Name
phone	Phone Number
address	Address
status	Agent Status

12. DATA FLOW DIAGRAM

Level 0 DFD:

Customer / Admin → Courier Management System → Database

The customer provides booking and tracking requests. The administrator manages customers, shipments, delivery agents, and status updates. The database stores all system information.

13. USE CASES

Customer: Register, Login, Book Courier, Track Courier, View Status.

Admin: Login, Manage Customers, Manage Shipments, Manage Delivery Agents, Update Status, View Reports.

Delivery Agent: Login, View Assigned Courier, Update Status, Confirm Delivery.

14. WORKING PROCESS

1. Customer registers in the system.
2. Customer logs in.
3. Customer enters sender and receiver details.
4. Customer books the courier.
5. System generates a tracking ID.
6. Admin verifies and manages the booking.
7. Courier is assigned to a delivery agent.
8. Delivery agent picks up the parcel.
9. Shipment status is updated.
10. Customer tracks the shipment.
11. Parcel is delivered to the receiver.
12. Delivery status is changed to Delivered.

15. TESTING

Test Case	Expected Result
Customer Registration	Account should be created
Customer Login	User should login successfully
Courier Booking	Booking should be stored
Tracking ID	Unique tracking ID should be generated
Shipment Tracking	Current status should be displayed
Admin Login	Admin dashboard should open
Agent Login	Agent dashboard should open

16. FUTURE ENHANCEMENTS

- Live GPS tracking
- Online payment
- SMS/Email notifications
- Mobile application
- Automatic delivery-agent assignment
- Route optimization
- Customer rating and review
- Barcode/QR-code tracking
- AI-based delivery management

17. CONCLUSION

The Courier Management System provides an efficient and systematic way to manage courier operations. It simplifies customer registration, courier booking, shipment management, delivery-agent assignment, and shipment tracking.

The system reduces manual work, minimizes errors, improves information management, and provides customers with an easy way to monitor their courier status. Therefore, the system can help courier organizations manage their delivery operations more effectively.

18. BIBLIOGRAPHY

- PHP Documentation
- MySQL Documentation
- HTML and CSS References
- JavaScript Documentation
- Database Management System textbooks
- Web Application Development textbooks

PROJECT SCREENSHOTS / EVIDENCE

Courier Management System (CMS)
Team Members: Raguvaran, Priya, Sabarinathan, Sanjeev, Rajeskumar

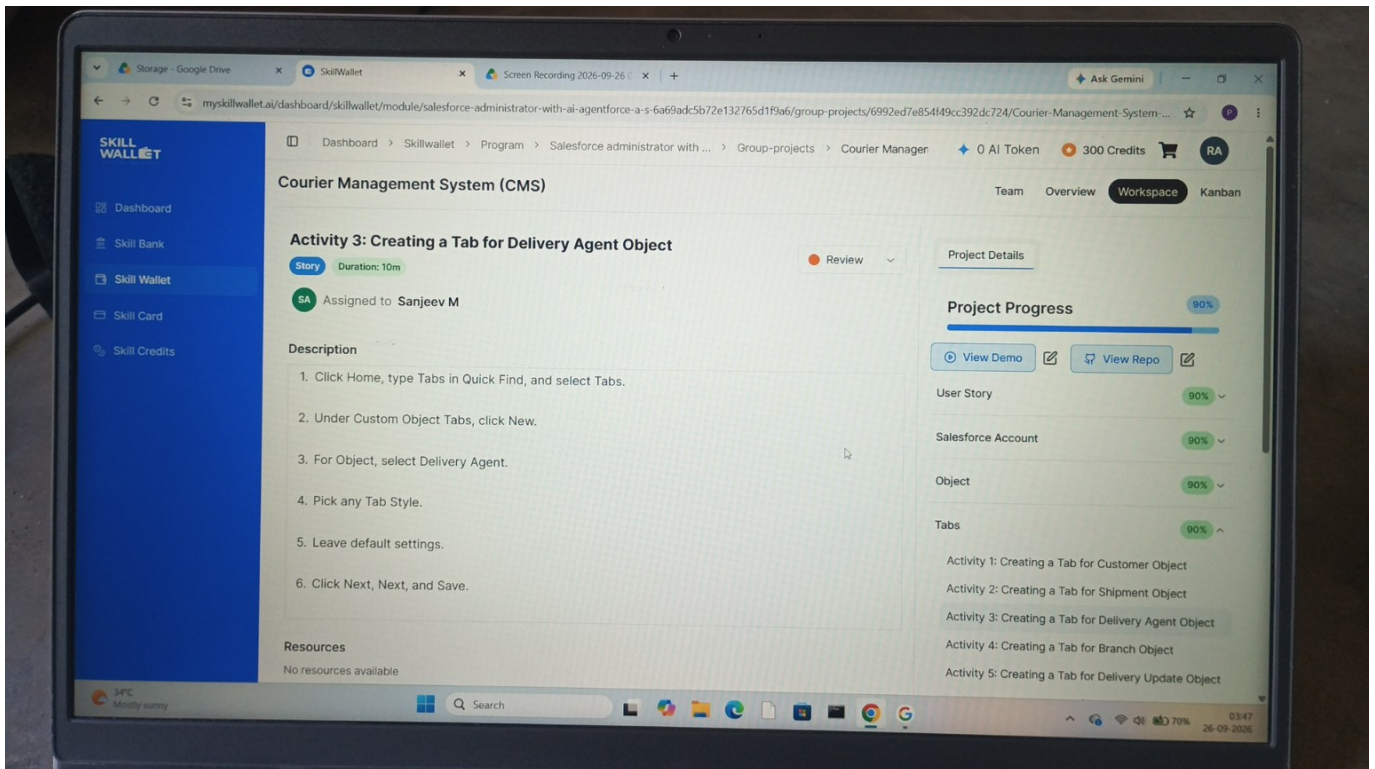


Figure 1: Activity 3 – Creating a Tab for Delivery Agent Object

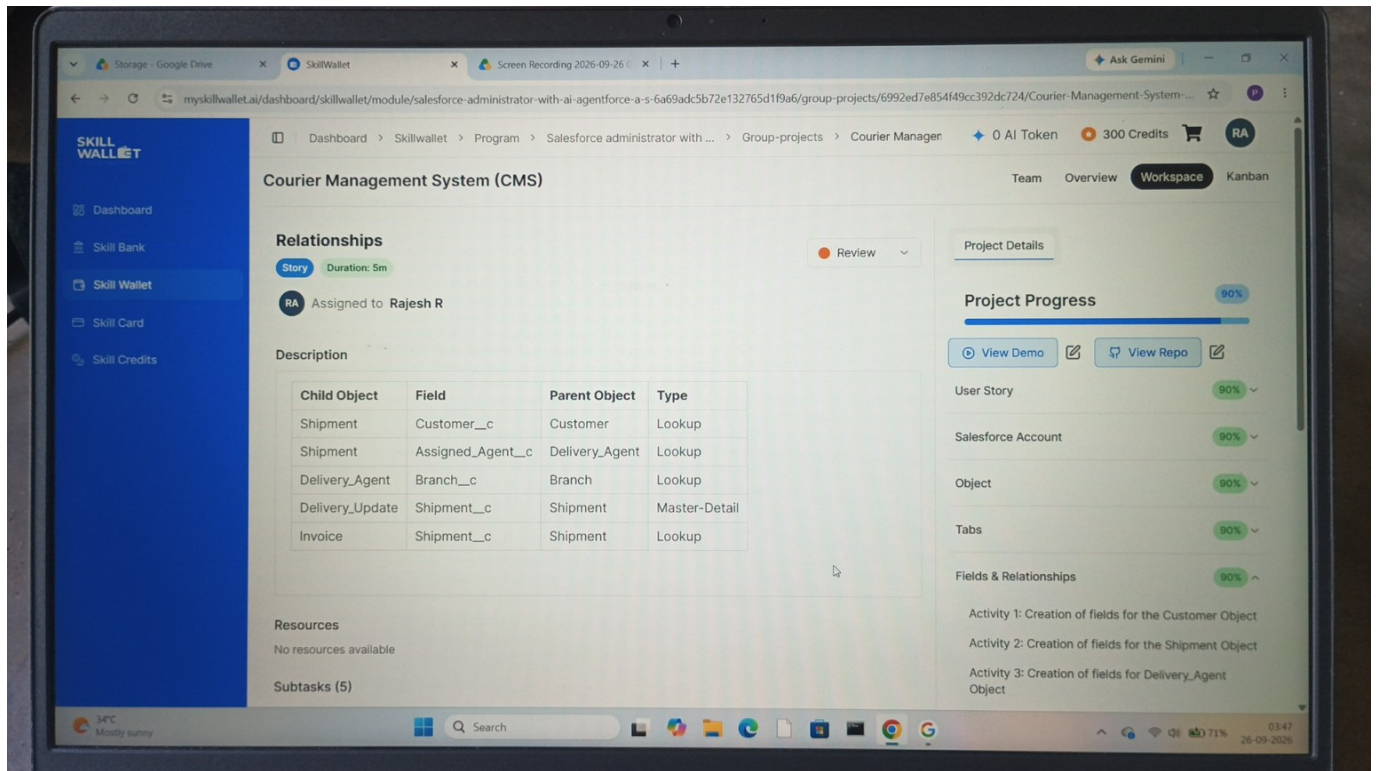


Figure 2: Relationships between Courier Management System Objects

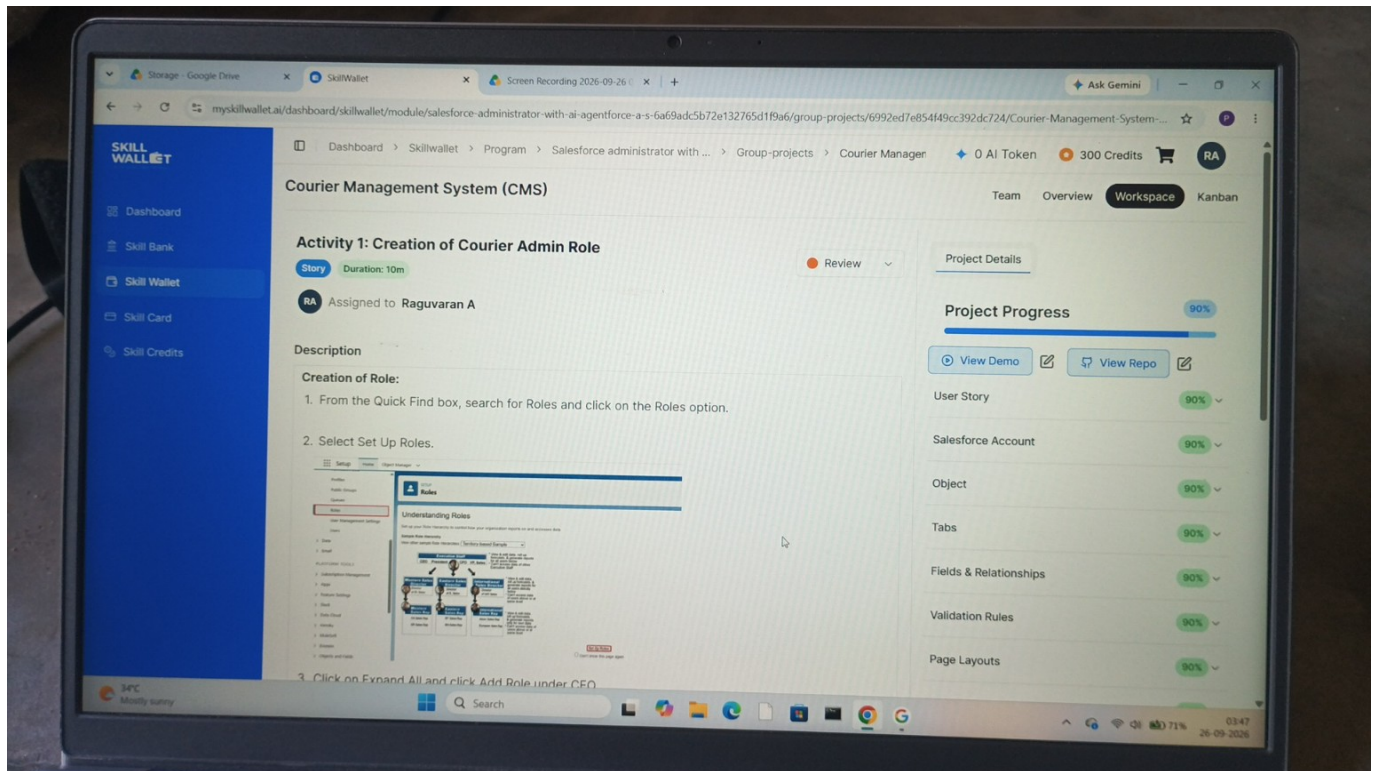


Figure 3: Activity 1 – Creation of Courier Admin Role

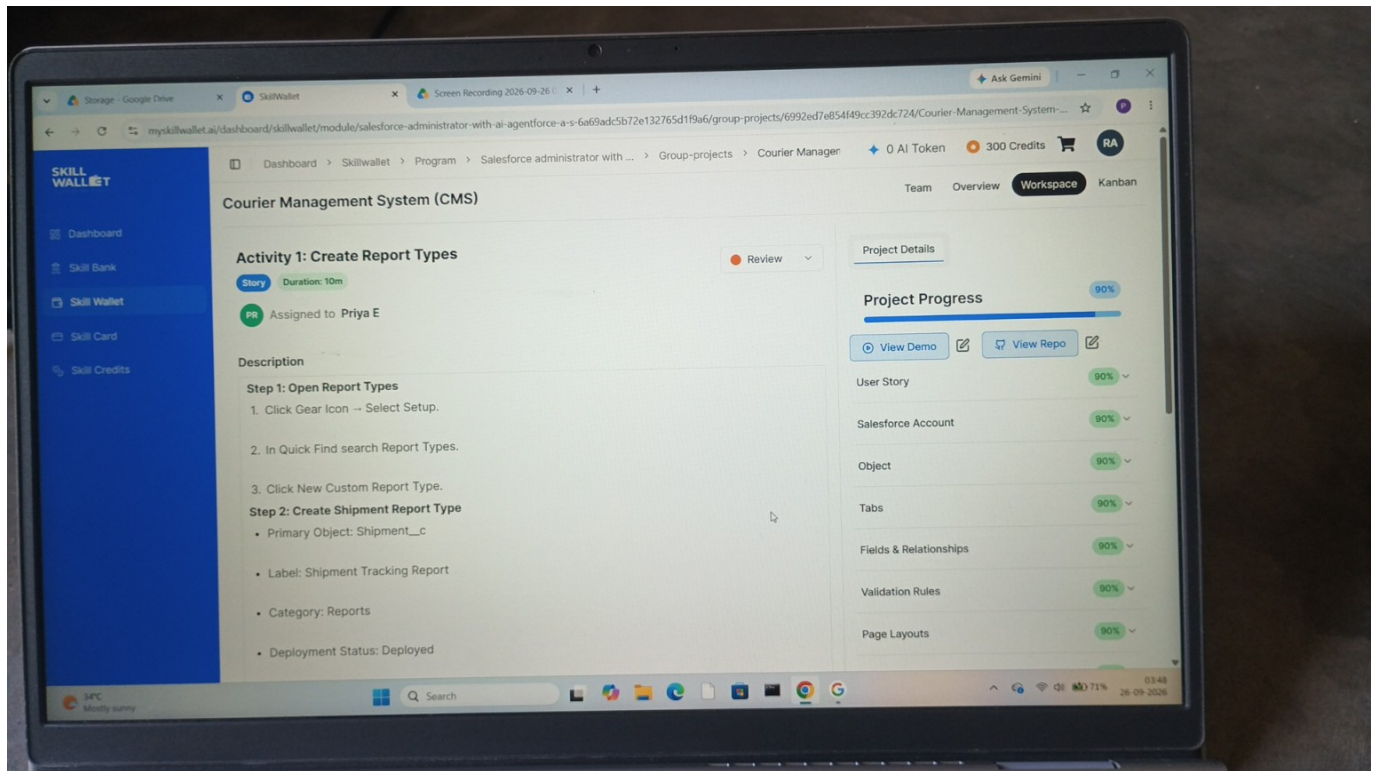


Figure 4: Activity 1 – Create Report Types / Shipment Tracking Report

PROJECT SCREENSHOTS / EVIDENCE

Figure 5: Activity 1 – Creating Developer Account

The screenshot displays the SkillWallet dashboard for a project titled "Courier Management System (CMS)". The left sidebar contains navigation links: Dashboard, Skill Bank, Skill Wallet, Skill Card, and Skill Credits. The main content area is titled "Activity 1: Creating Developer Account" and includes a "Story" tab, a duration of "30m", and a "Review" button. The activity is assigned to "Sabarinathan K". The description states: "Creating a developer org in salesforce." and lists four steps: 1. Go to <https://developer.salesforce.com/signup>, 2. On the sign up form, enter the following details :, 3. First name & Last name, 4. Email. Below the text is a screenshot of the Salesforce Developer Edition sign-up form. The right sidebar shows "Project Progress" at 90%, with a "View Demo" and "View Repo" button. Below this is a list of project components with their progress: User Story (90%), Salesforce Account (90%), Activity 1: Creating Developer Account, Activity 2: Account Activation, Object (90%), Tabs (90%), Fields & Relationships (90%), and Validation Rules (90%). The bottom of the screen shows a Windows taskbar with the date 26-09-2026 and time 03:46.

Courier Management System (CMS)

Team Overview **Workspace** Kanban

Activity 1: Creating Developer Account

Story Duration: 30m **Review**

SA Assigned to Sabarinathan K

Description

Creating a developer org in salesforce.

1. Go to <https://developer.salesforce.com/signup>
2. On the sign up form, enter the following details :
3. First name & Last name
4. Email

Project Progress 90%

[View Demo](#) [View Repo](#)

User Story 90%

Salesforce Account 90%

Activity 1: Creating Developer Account

Activity 2: Account Activation

Object 90%

Tabs 90%

Fields & Relationships 90%

Validation Rules 90%